

| Time \ Day             | June 22<br>Monday                                                                                                                        | June 23<br>Tuesday                                                                                                                                        | June 24<br>Wednesday                                                                                              | June 25<br>Thursday                                                                                                        | June 26<br>Friday                                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 8:30-8:50              | Registration                                                                                                                             |                                                                                                                                                           |                                                                                                                   |                                                                                                                            |                                                                                                                            |
| 8:50-9:00              | Opening                                                                                                                                  |                                                                                                                                                           |                                                                                                                   |                                                                                                                            |                                                                                                                            |
| Morning<br>Session 1   | Chair: Masahito Hayashi                                                                                                                  | Chair: Xin Wang                                                                                                                                           | Chair: Nelly Ng Huei<br>Ying                                                                                      | Chair: Ge Bai                                                                                                              | Chair: Hao-Chung Cheng                                                                                                     |
| 9:00-9:30              | <b>Ludovico Lami:</b><br>On some extensions of the<br>generalised quantum Stein<br>'s lemma                                              | <b>Benjamin Lovitz:</b><br>Constructive<br>counterexamples to the<br>additivity of minimum output<br>Rényi entropy of quantum<br>channels for all $p > 1$ | <b>Marco Tomamichel:</b><br>A complete<br>characterisation of<br>conditional entropies                            | <b>Aby Philip:</b><br>Robustness of quantum<br>data hiding against<br>entangled catalysts and<br>memory                    | <b>Amin Aminzadeh Gohari:</b><br>The auxiliary receiver<br>approach in network<br>information theory                       |
| 9:30-10:00             |                                                                                                                                          |                                                                                                                                                           | <b>Yunlong Xiao:</b><br>No practical quantum<br>broadcasting: even<br>virtually                                   | <b>Hailey Murray:</b><br>Fundamental limits for<br>thermodynamic control<br>with quantum feedback<br>(online Q&A)          |                                                                                                                            |
| 10:00-10:45            | Coffee Break                                                                                                                             | Coffee Break                                                                                                                                              | Coffee Break<br>& Group Photo                                                                                     | Coffee Break                                                                                                               | Coffee Break                                                                                                               |
| Morning<br>Session 2   | Chair: Milad Moazami<br>Goodarzi                                                                                                         | Chair: Benjamin Lovitz                                                                                                                                    | Chair: Ian Thomas<br>George                                                                                       | Chair: Haidong Yuan                                                                                                        | Chair: Aby Philip                                                                                                          |
| 10:45-11:15            | <b>Bartosz Regula:</b><br>Beyond Hoeffding and<br>Chernoff: trading<br>conclusiveness for<br>advantages in quantum<br>hypothesis testing | <b>Kaito Watanabe:</b><br>Reliability function of work<br>extraction and beyond                                                                           | <b>Pablo Costa Rico:</b><br>Partial trace relations<br>beyond normal matrices                                     | <b>Idris Natanaël Delsol:</b><br>Computational complexity<br>of optimally encoding a<br>qubit                              | <b>Martin Sandfuchs:</b><br>Randomness from<br>causally independent<br>processes                                           |
| 11:15-11:45            | <b>Kun Fang:</b><br>An almost complete<br>picture of quantum state<br>discrimination with<br>composite correlated<br>hypotheses          | <b>Milad Moazami Goodarzi:</b><br>Sandwiched Rényi<br>divergence and the<br>quantum central limit<br>theorem                                              | <b>Afham:</b><br>Projections with respect to<br>bures distance and fidelity:<br>closed-forms and<br>applications  | <b>Pablo Costa Rico:</b><br>Sharp inequalities for<br>Schur-convex functionals<br>of partial traces over<br>unitary orbits | <b>Filippo Girardi:</b><br>Random purification<br>channel made simple                                                      |
| 11:45-13:30            | Lunch Break (Buffet)                                                                                                                     | Lunch Break (Buffet)                                                                                                                                      | Lunch Break (Buffet)                                                                                              | Lunch Break (Buffet)                                                                                                       | Lunch Break (Buffet)                                                                                                       |
| Afternoon<br>Session 1 | Chair: Bartosz Regula                                                                                                                    | Chair: Afham                                                                                                                                              | Chair: Masahito Hayashi                                                                                           | Chair: Ludovico Lami                                                                                                       | Chair: Ranyiliu Chen                                                                                                       |
| 13:30-14:00            | <b>Sujeet Bhalerao:</b><br>Improving quantum<br>communication rates with<br>permutation-invariant<br>codes                               | <b>Bartosz Regula:</b><br>Rethinking quantum smooth<br>entropies: tight one-shot<br>analysis of quantum privacy<br>amplification                          | <b>Alexander Holevo:</b><br>Convex closure of the<br>output entropy and sharp<br>entropy inequalities<br>(online) | <b>Salman Beigi:</b><br>Fermions are<br>fundamentally more<br>nonlocal than Bosons<br>(online)                             | <b>Arick Grootveld:</b><br>The rate function for<br>quantum state tomography                                               |
| 14:00-14:30            | <b>Idris Natanaël Delsol:</b><br>Emulation capacity<br>between idempotent<br>channels                                                    | <b>Yongsheng Yao:</b><br>Tight any-shot quantum<br>decoupling                                                                                             |                                                                                                                   |                                                                                                                            | <b>Qiushi Liu:</b><br>Subsystem quantum error<br>correction for noisy<br>quantum metrology                                 |
| 14:30-15:00            | Coffee Break                                                                                                                             | Coffee Break                                                                                                                                              | Free Time                                                                                                         | Coffee Break                                                                                                               | Coffee Break                                                                                                               |
| Afternoon<br>Session 2 | Chair: Matt Hoogsteder-<br>Riera                                                                                                         | Chair: Ryuji Takagi                                                                                                                                       |                                                                                                                   | Chair: Pablo Costa Rico                                                                                                    | Chair: Kun Fang                                                                                                            |
| 15:00-15:30            | <b>Liuhan Ye:</b><br>Strong converse bounds<br>on the classical<br>identification capacity of<br>the qubit depolarizing<br>channel       | <b>Marco Tomamichel:</b><br>The strong converse<br>exponent of composable<br>randomness extraction<br>against quantum side<br>information                 |                                                                                                                   | <b>Benchi Zhao:</b><br>Power and limitations of<br>distributed quantum state<br>purification                               | <b>Sang-Jun Park:</b><br>k-Positivity and high-<br>dimensional bound<br>entanglement under<br>symplectic group<br>symmetry |
| 15:30-16:00            | <b>Xuanqiang Zhao:</b><br>The communication power<br>of indefinite causal order                                                          | <b>Masahito Hayashi:</b><br>Universal classical-quantum<br>channel resolvability and its<br>application to universal<br>private-channel coding            |                                                                                                                   | <b>Ludovico Lami:</b><br>Fundamental quality bound<br>on optical quantum<br>communication                                  | <b>Christopher Vairags:</b><br>Exact finite group designs<br>under symmetry                                                |
| 16:00-18:00            | Poster Session                                                                                                                           | Open Problem Session<br>(Chair: Hao-Chung<br>Cheng)                                                                                                       |                                                                                                                   | Free Time                                                                                                                  | Closing<br>(16:00-16:15)                                                                                                   |
| 18:00-21:30            | Free Time                                                                                                                                | Free Time                                                                                                                                                 | Banquet (Buffet)<br>in COLI hotel                                                                                 | Free Time                                                                                                                  |                                                                                                                            |